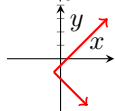


You must show your work to receive credit. Unless stated otherwise, all problems are worth 5 points.

Determine whether the following relation defines y as a function of x .

1. $x = |y + 3| - 1$



Find the domain and range of the relation. Determine if the relation is a function.

2. $g = \{(2, 7), (30, -4), (45, 7), (2, 5)\}$

a. Domain:

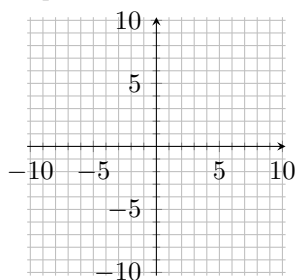
b. Range:

c. Is g a function?

Sketch the graph of the function, then give its domain and range.

3. $y = -2x + 4$

a. Graph:

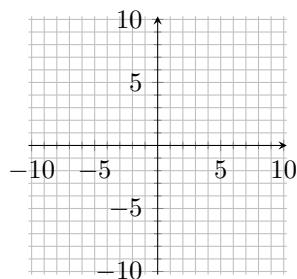


b. Domain:

c. Range:

4. $f(x) = \begin{cases} x + 2 & \text{for } x > 1 \\ -x - 4 & \text{for } x \leq 1 \end{cases}$

a. Graph:



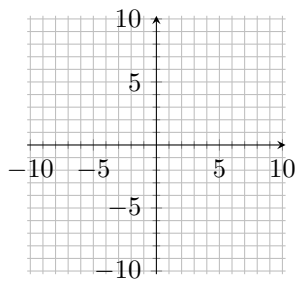
b. Domain:

c. Range:

Use transformation to graph the function. State the domain and range using interval notation.

5. $f(x) = -(x + 3)^2 + 5$

a. Graph:

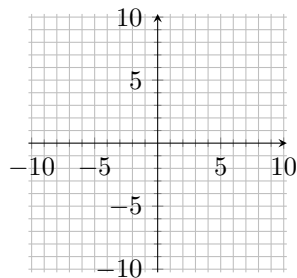


b. Domain:

c. Range:

6. $g(x) = -\frac{1}{2}|x - 1|$

a. Graph:

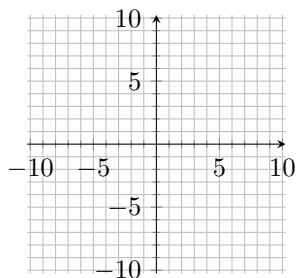


b. Is $g(x)$ one-to-one? i.e., does it have an inverse?

Sketch the graph and state the domain and range. Identify any intervals on which $f(x)$ is increasing or decreasing.

7. $f(x) = -\sqrt{x-3}$

a. Graph:



b. Increasing:

c. Decreasing:

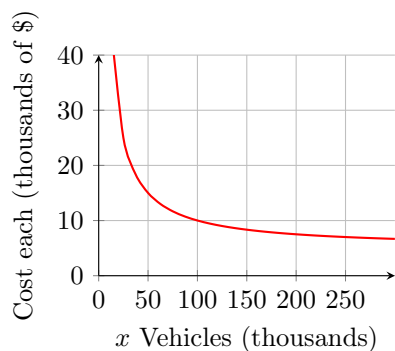
Use the function $h(x) = 4x + 5$ to complete the following problems.

8. Find $h(4)$

9. $h(0) = 5$. Find $h^{-1}(5)$

Use the provide graph to answer the question.

10. The manufacturing cost per vehicle for production of x vehicles is shown in the graph. Use the graph to find the average rate of change of the cost per vehicle from 50 to 100 thousand vehicles.



Use the difference quotient, $\frac{f(x+h) - f(x)}{h}$ for $h \neq 0$, to complete and simplify the following problem.

11. $f(x) = 7x + 3$.

Find the inverse of the following function.

12. $f(x) = -\frac{1}{3}x + 1$

Let $f(x) = x + 4$ and $g(x) = x^2 - x$. Find and simplify the expression.

13. $(f + g)(-6)$

14. $(g \circ f)(2)$

For the given functions, find $(f \circ g)(x)$ and $(g \circ f)(x)$, Determine if g is the inverse of f .

15. $f(x) = x^2 + 1$ and $g(x) = \sqrt{x}$.

a. $(f \circ g)(x)$

b. $(g \circ f)(x)$

c. Is g the inverse of f ?